Ahsan Habib Akash

Ph.D. Student in Electrical Engineering · Mechanistic Interpretability & Evaluation of Vision–Language Models
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Research Interests

Mechanistic interpretability, evaluation, and post-training of vision–language models (VLMs): whether reinforcement-learning post-training yields genuine visual grounding or amplifies language priors; causal analysis of how VLMs bind visual evidence to answers; diagnostic benchmarks that expose failures hidden by aggregate metrics; and trustworthy, fair VLMs for medical imaging.

Education

West Virginia University — Morgantown, WV
Ph.D. in Electrical Engineering · Specialization: Computer Vision — Jan 2024 – Dec 2028 (expected)
Advisor: Dr. Prashnna K. Gyawali
M.S. in Electrical Engineering (earned en route to Ph.D.) — Summer 2026
Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
B.Sc. in Electrical and Electronic Engineering — Feb 2017 – May 2022

Publications

Conference & Workshop Papers

- **When Medical VLMs Stop Understanding: MedTEC-Bench for Probing Semantic Specificity** 2026
  Ahsan Habib Akash, Alina Devkota, Donald Adjeroh, Binod Bhattarai, Prashnna K. Gyawali
  Advances in Neural Information Processing Systems (NeurIPS 2026), Evaluations & Datasets Track · Poster
- **Two is Better than One: A Collapse-free Multi-Reward RLIF Training Framework** 2026
  Shourov Joarder, Diganta Sikdar, **Ahsan Habib Akash**, Binod Bhattarai, Prashnna K. Gyawali
  Advances in Neural Information Processing Systems (NeurIPS 2026), Main Track · Poster · arXiv:2605.22620
- **Addressing Bias in VLMs for Glaucoma Detection Without Protected Attribute Supervision** 2025
  Ahsan Habib Akash, Greg Murray, Annahita Amireskandari, Joel Palko, Carol Laxson, Binod Bhattarai, Prashnna K. Gyawali
  3rd Workshop on Data Engineering in Medical Imaging (DEMI), MICCAI 2025 · arXiv:2508.09087
- **Study of Multi-class Arrhythmia Detection and Classification from ECG Data using Deep Learning** 2023
  Ahsan Habib Akash, R. Ahamed
  26th Int. Conf. on Computer and Information Technology (ICCIT) · doi:10.1109/ICCIT60459.2023.10441043
- **SignBD-Word: Video-Based Bangla Word-Level Sign Language and Pose Translation** 2023
  A. Sams, **Ahsan Habib Akash**, S. M. M. Rahman
  14th Int. Conf. on Computing, Communication and Networking Technologies (ICCCNT) · doi:10.1109/ICCCNT56998.2023.10306914
- **Activity Classification from First-Person Office Videos with Visual Privacy Protection** 2022
  Partho Ghosh, Md Abrar Istiak, Nayeeb Rashid, **Ahsan Habib Akash**, Ridwan Abrar, Ankan Ghosh Dastider, Asif Shahriyar Sushmit, Taufiq Hasan
  The Fourth Industrial Revolution and Beyond, Springer Nature, pp. 157–169

Journal Articles

- **Robust CNN Multi-Nested-LSTM Framework with Compound Loss for Patch-Based Multi-Push Ultrasound Shear Wave Imaging and Segmentation** 2026
  Md. Jahin Alam, **Ahsan Habib Akash**, Muyinatu A. Lediju Bell, Md. Kamrul Hasan
  Physics in Medicine & Biology, Early Career Researcher Focus Collection · doi:10.1088/1361-6560/ae2db8

Under Review & Preprints

- **Does RL Post-Training Ground Vision–Language Models? A Causal Measurement** 2026
  Ahsan Habib Akash et al.
  Under review at ICLR 2027
- **LAD: Language-Anchored Decomposition for Concept Discovery** 2026
  Ahsan Habib Akash et al.
  Preprint · code and project page
- **SW-ViT: A Spatio-Temporal Vision Transformer Network with Post Denoiser for Sequential Multi-Push Ultrasound Shear Wave Elastography** 2025
  Ahsan Habib Akash, Md. Jahin Alam, Md. Kamrul Hasan
  arXiv:2505.18865

Research Experience

West Virginia University — Morgantown, WV
Graduate Research Assistant · Advisor: Dr. Prashnna K. Gyawali — Jan 2025 – Present

- **Causal analysis of RL post-training in VLMs:** built a counterfactual benchmark with solver-certified edits and an activation-patching pipeline, and tested GRPO, VPPO, and PRPO on four Qwen-VL backbones. Found that RL raises edit following modestly, yet models keep the pre-edit answer after referent edits (prior lock) and the gain is only partly carried at the evidence location (first author, under review). Current work localizes the binding mechanism and tests targeted inference-time fixes.
- **Evaluation & benchmarking:** built MedTEC-Bench, a diagnostic benchmark with a five-level probing taxonomy across 11 models and 6 imaging modalities, exposing semantic-specificity failures in medical VLMs hidden by aggregate accuracy (first author, NeurIPS 2026 E&D).
- **Unsupervised RL post-training of LLMs:** co-developed a collapse-free multi-reward reinforcement-learning-from-internal-feedback (RLIF) framework that pairs answer-level cluster-voting and completion-level self-certainty rewards, with GDPO per-reward normalization and KL-Cov regularization on entropy-collapsing tokens. It retains 89–99% of peak accuracy after 340 training steps on Qwen2.5-1.5B/3B, where single-reward RLIF collapses to near zero, and tracks supervised RLVR without any ground-truth labels (NeurIPS 2026).
- **Interpretable & fair medical AI:** language-anchored concept discovery (LAD; NMF with CLIP-guided similarity maps) and sparse autoencoders for human-interpretable concept directions; unsupervised bias mitigation for VLM-based glaucoma screening without protected-attribute labels (DEMI @ MICCAI 2025).

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
Research Assistant, DSP Research Lab · Advisor: Dr. Md. Kamrul Hasan — Dec 2022 – Dec 2023

- Collaborated with Fujifilm (Japan and USA) on deep learning for ultrasound shear-wave elastography acquisition.
- Built multi-resolution spatio-temporal vision-transformer models (SW-ViT) with a synthetic-to-real pipeline: COMSOL Multiphysics simulation for pre-training, CIRS phantom data for fine-tuning.
- Led medical image segmentation and classification projects, resulting in a first-author preprint and a co-authored journal article (Physics in Medicine & Biology).

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
Research Assistant, Telecommunication Lab · Advisor: Dr. Md. Farhad Hossain — Sep 2022 – Dec 2022

- Developed deep-learning models for anomaly detection in network traffic.

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
Undergraduate Research · Advisor: Dr. S. M. Mahbubur Rahman — Aug 2021 – 2022

- Video-based Bangla sign-language translation (SignBD-Word) with I3D, SlowFast, and CNN–LSTM recognition models (ICCCNT 2023).

Awards & Achievements

2nd place, MICCAI 2025 Multi-Camera Robust Diagnosis of Fundus Diseases (MuCaRD) Challenge	2025
Top 8, AI for Bangla 2.0	2023
Top 6, Robi Datathon 2.0	2022
1st place, IEEE VIP Cup @ ICIP 2020: real-time vehicle detection and tracking with fisheye cameras	2020
5th place, IEEE Signal Processing Cup @ ICASSP 2020: real-time anomaly detection in ground/aerial systems	2020
3rd place, IEEE VIP Cup @ ICIP 2019: activity recognition from body cameras	2019

Technical Skills

ML research	Mechanistic interpretability (activation patching, causal tracing, sparse autoencoders), RL post-training (GRPO, RLIF), benchmark design and evaluation, bias mitigation, federated learning
Languages & tools	Python, PyTorch, TransformerLens, EasyR1, TensorFlow, OpenCV, C/C++, MATLAB, COMSOL Multiphysics
Compute & allocations	HPC clusters (PSC Bridges-2, NCSA DeltaAI, TACC Vista, CSCS Clariden); NAIRR Pilot Start-Up NAIRR260295 (TACC Vista); ACCESS CIS261254

Selected Course Projects

West Virginia University, 2024 – present: multi-stage GAN for satellite image super-resolution · residual diffusion denoising for universal image restoration · attention graph neural network for word-level ASL recognition · windowed graph attention for image compression and super-resolution.